Listing 4.2 `describe_image.py`

```
import base64
import requests
import os

def describe_image(image_path='animals.png') -> str:
    """
    Uses GPT-4 Vision to inspect and describe the contents of the image.

    :param input_path: str, the name of the PNG file to describe.
    """
    api_key = os.environ['OPEN_API_KEY']

    # Function to encode the image
    def encode_image(image_path):      #1
        with open(image_path, "rb") as image_file:
            return base64.b64encode(image_file.read()).decode('utf-8')
    # Getting the base64 string
    base64_image = encode_image(image_path)

    headers = {
    "Content-Type": "application/json",
    "Authorization": f"Bearer {api_key}"
    }

    payload = {
    "model": "gpt-4-turbo",
    "messages": [
        {
        "role": "user",
        "content": [
            {
            "type": "text",
            "text": "What's in this image?"
            },
            {
            "type": "image_url",
            "image_url": {
         "url": f"{base64_image}"      #2
            }
            }
        ]
        }
    ],
    "max_tokens": 300
    }

    response = requests.post(
        "https://api.openai.com/v1/chat/completions",
        headers=headers,
        json=payload)

    return response.json()["choices"][0]["message"]  #3
["content"]                                          #3
```

#1 Function to load and encode the image as a Base64 string
#2 Including the image string along with the JSON payload
#3 Unpacking the response and returning the content of the reply